MACHINE LEARNING

Feature Engineering Visual Guide

The highest-ROI step in ML:
encoding, scaling, selection, leakage traps.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Encoding Categoricals

Turn labels into numbers without inventing order.

One-hot

binary columns.

Ordinal

ordered integers.

Target

category with mean target value.

cross-validation!

Label-encoding 'red'=1,'blue'=2' invents an order the model will believe. High-cardinality IDs? Hash them or embed them.

STEP 2/5

Scaling

Put features on comparable footing.

Standardize

zero mean, unit variance.
For gradient methods, SVMs.

(x - mean) / std

Normalize

range [0,1]. For neural nets, kNN, images.

(x - min) / (max - min)

Fit scalers on TRAIN only, then apply to test — fitting on all data leaks test statistics into training.

STEP 3/5

Feature Selection

Fewer, better features beat more features.

- Drop constant and near-constant features first — they carry no signal.
- Drop one of each highly-correlated pair; trees don't care, linear models do.
- L1 regularization selects features automatically during training.

STEP 4/5

Target Leakage Traps

The silent killer of ML projects.

Leak example

predicting readmission — 'days_in_hospital' known only AFTER. –

Leak example

loan data — recorded 'approved' flag in after the decision.

Leakage test: could this feature exist at prediction time in production? If not, drop it. Great validation scores + leakage = worthless model.

STEP 5/5

Feature Engineering Workflow

A repeatable recipe.

- Start with domain features: ratios, differences, and aggregates a human expert would check.
- Add time features: hour, weekday, days-since-event, rolling averages with proper cutoffs.
- Create interactions the model can't easily learn: price_per_sqft beats price + sqft for linear models.
- Build everything inside a pipeline (fit on train, transform both) so leakage can't creep in.
- Version features in a feature store — training/serving skew starts as a notebook shortcut.

STEP ✓I✓

Key Takeaways

Feature Engineering — Remember

- One-hot nominal categories; never invent order with label encoding.
- Scale for gradient/kNN/neural models; fit scalers on train only.
- Select features: drop constants, dedupe correlates, let L1 help.
- Leakage = features unavailable at prediction time; it fakes great scores.
- Pipelines + feature stores keep train/serve consistent and leakage out.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev